

$$, \quad \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{(1-\bar{\alpha}_t)} \mathbf{I} \Big)$$

$c'_t \rightsquigarrow$

$$D_{KL} (q(x_{t-1} | x_t, x_0) \parallel \underbrace{P_\theta(x_{t-1} | x_t)}_{\mathcal{N}(\mu_\theta(x_t, t), \Sigma_3)})$$

$$= \frac{\|\mu_3 - \mu_\theta\|^2}{2\sigma_3^2} = \left\| \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t} \underline{x_0} - \underbrace{\mu_\theta(x_t, t)}_{\in \mathcal{N}_t + \epsilon_t(x_t, t)} \right\|_2^2 / 2 \left(\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \right)$$

بدون

Try to write x_0 in terms of x_t .

$$x_t = x_{t-1} \sqrt{1-\beta_t} + \sqrt{\beta_t} \epsilon_t$$

$$x_t = \sqrt{\alpha_t \alpha_{t-1} \dots \alpha_0} x_0 + \sqrt{1-\alpha_t \alpha_{t-1} \dots \alpha_0} \epsilon_t \quad \epsilon_t \sim \mathcal{N}(0, \mathbf{I})$$

$$= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} \epsilon_t$$

$$\Rightarrow x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} \epsilon_t$$

$$\begin{aligned} \text{t-th loss term} &= \left\| \frac{\sqrt{\bar{\alpha}_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t} \left[\frac{1}{\sqrt{\bar{\alpha}_t}} x_t - \frac{\sqrt{1-\bar{\alpha}_t} \epsilon_t}{\sqrt{\bar{\alpha}_t}} \right] - \mu_\theta(x_t, t) \right\|_2^2 / c \end{aligned}$$

$$\begin{aligned}
&= \left\| \frac{\sqrt{\bar{\alpha}_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t + \frac{1-\alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} x_t - \frac{(1-\alpha_t)}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \epsilon_t \right\|^2 \\
&= \left\| \frac{\sqrt{\alpha_t}\sqrt{\bar{\alpha}_t}(1-\bar{\alpha}_{t-1}) + 1-\alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} x_t - \frac{(1-\alpha_t)}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \epsilon_t \right\|^2 \\
&\quad \begin{array}{c} \gamma_t \sim (1-\bar{\alpha}_t)\sqrt{\alpha_t} \\ \mu := \gamma_t x_t - \xi_t \epsilon_\theta(x_t, t) \end{array} \quad \begin{array}{c} \gamma_t x_t - \xi_t \epsilon_t \\ \boxed{\gamma_\theta} \\ \uparrow \\ x_t \end{array} \quad \xi_t \leftarrow \frac{(1-\alpha_t)}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \epsilon_t \\
&= \left\| -\xi_t \epsilon_t + \xi_t \epsilon_\theta(x_t, t) \right\|_2^2 / c = \underbrace{\xi_t^2 / c} \underbrace{\left\| \epsilon_t - \epsilon_\theta(x_t, t) \right\|_2^2}
\end{aligned}$$

$$\text{Modified ELBO} = \mathbb{E}_{t \sim U(0, T)} \left\{ \left\| \epsilon_t - \epsilon_\theta(x_t, t) \right\|_2^2 \right\}$$

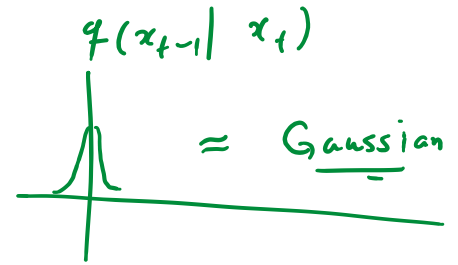
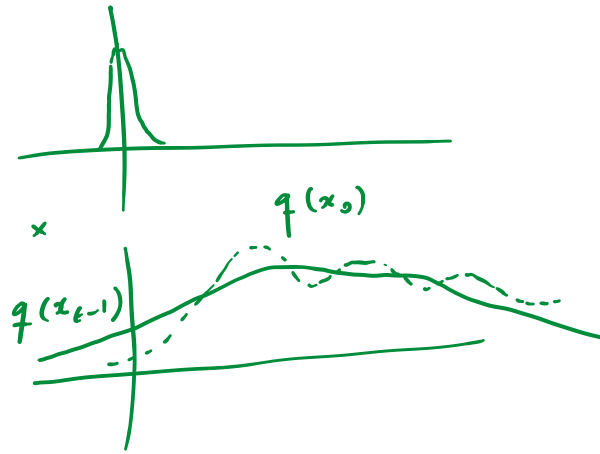
Training Recipe:

- ① Take n sample $x_0^{(1)} \dots x_0^{(n)}$.
- ② Take n time step $\overset{\text{iid}}{\sim} U(1, T)$, $t_1 \dots t_n$
- ③ $\forall i$ $x_{t_i}^{(i)} = x_0^{(i)} \sqrt{\bar{\alpha}_{t_i}} + \sqrt{1-\bar{\alpha}_{t_i}} \epsilon^{(i)}$ $\epsilon^{(i)} \sim \mathcal{N}(0, I)$
- ④ Take loss as $\sum_{i=1}^n \left\| \epsilon^{(i)} - \epsilon_\theta(x_{t_i}^{(i)}, t_i) \right\|_2^2$.
- ⑤ Take $\nabla_\theta \mathcal{L}$ & update θ in the neg. dir. of ∇ grad.

Inference Recipe:

- ① $x_T \sim \mathcal{N}(0, I)$ (Langevin Dynamics)
- ② $\forall t \in \{0, \dots, T-1\}$ $x_{t-1} = \gamma_t x_t - \xi_t \epsilon_\theta(x_t, t) + \zeta_t z_t$
(Iterate) $z_t \sim \mathcal{N}(0, I)$

$$q(x_{t-1} | x_t) = \frac{q(x_t | x_{t-1}) q(x_{t-1})}{q(x_t)}$$



$$\approx P_\theta(x_{t-1} | x_t) = \mathcal{N}(\underbrace{\mu_\theta(x_t, t)}_{\gamma_t x_t - \sum_t \epsilon_t}, c_t' I)$$

$$d_t \{ \epsilon_1 - \epsilon_\theta(x_1, 1) \}$$

$$\mathbb{E}_{\substack{\{ \log P_\theta(x_0 | x_1) \} \\ q(x_1 | x_0)}} = \mathbb{E}_{\substack{\leftarrow \\ q(x_1 | x_0)}} \left\{ - \frac{\| x_0 - \mu(x_1, 1) \|^2}{2\sigma^2} \right\} + c''$$